

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

“JnanaSangama”, Belgaum- 590014, Karnataka.



LAB RECORD

on

Big Data Analytics (23CS6PCBDA)

Submitted by

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in partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING



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Bull Temple Road, Bangalore 560019
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Department of Computer Science and Engineering



CERTIFICATE

This is to certify that the Lab work entitled “Big Data Analytics” carried out by **Dhruthi Vijay (1BM23CS369)**, who is bonafide student of **B.M.S. College of Engineering**. It is in partial fulfilment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum during the year 2026. The Lab report has been approved as it satisfies the academic requirements in respect of a Big Data Analytics (23CS6PCBDA) work prescribed for the said degree.

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Github Link: <https://github.com/DhruthiVijay/BDA-LAB>

Course Outcomes (COs):

CO1	Apply the concept of NoSQL, Hadoop or Spark for a given task
CO2	Analyse big data analytics mechanisms that can be applied to obtain solution for a given problem.
CO3	Design and implement solutions using data analytics mechanisms for a given problem.

PROGRAM – 1

MongoDB- CRUD Operations Demonstration (Practice and Self Study)

Commands with output:

Student database

1. Create a database “Student” with the following attributes Rollno, Age, ContactNo, Email-Id.

```
Atlas atlas-b6pfyk-shard-0 [primary] test> db.createCollection("Student");
{ ok: 1 }
```

2. Insert appropriate values (at least 5)

```
Atlas atlas-b6pfyk-shard-0 [primary] test> db.Student.insert({RollNo:1, Age:21, Cont:9876, email:"antara.de9@gmail.com"});
DeprecationWarning: Collection.insert() is deprecated. Use insertOne, insertMany, or bulkWrite.
{
  acknowledged: true,
  insertedIds: { '0': ObjectId("660a82ec7c840f42b4a05530") }
}
Atlas atlas-b6pfyk-shard-0 [primary] test>

Atlas atlas-b6pfyk-shard-0 [primary] test> db.Student.insert({RollNo:2, Age:22, Cont:9976, email:"anushka.de9@gmail.com"});
{
  acknowledged: true,
  insertedIds: { '0': ObjectId("660a82ed7c840f42b4a05531") }
}
Atlas atlas-b6pfyk-shard-0 [primary] test>

Atlas atlas-b6pfyk-shard-0 [primary] test> db.Student.insert({RollNo:3, Age:21, Cont:5576, email:"anubhav.de9@gmail.com"});
{
  acknowledged: true,
  insertedIds: { '0': ObjectId("660a82ed7c840f42b4a05532") }
}
Atlas atlas-b6pfyk-shard-0 [primary] test>

Atlas atlas-b6pfyk-shard-0 [primary] test> db.Student.insert({RollNo:4, Age:20, Cont:4476, email:"pani.de9@gmail.com"});
{
  acknowledged: true,
  insertedIds: { '0': ObjectId("660a82ed7c840f42b4a05533") }
}
Atlas atlas-b6pfyk-shard-0 [primary] test>

Atlas atlas-b6pfyk-shard-0 [primary] test> db.Student.insert({RollNo:10, Age:23, Cont:2276, email:"rekha.de9@gmail.com"});
{
  acknowledged: true,
```

```
Atlas atlas-b6pfyk-shard-0 [primary] test> db.Student.insert({RollNo:10, Age:23, Cont:2276, email:"rekha.de9@gmail.com"});
{
  acknowledged: true,
  insertedIds: { '0': ObjectId("660a82f47c840f42b4a05534") }
}
```

3. View the data

```
Atlas atlas-b6pfyk-shard-0 [primary] test> db.Student.find()
[
  {
    _id: ObjectId("660a82ec7c840f42b4a05530"),
    RollNo: 1,
    Age: 21,
    Cont: 9876,
    email: 'antara.de9@gmail.com'
  },
  {
    _id: ObjectId("660a82ed7c840f42b4a05531"),
    RollNo: 2,
    Age: 22,
    Cont: 9976,
    email: 'anushka.de9@gmail.com'
  },
  {
    _id: ObjectId("660a82ed7c840f42b4a05532"),
    RollNo: 3,
    Age: 21,
    Cont: 5576,
    email: 'anubhav.de9@gmail.com'
  },
  {
    _id: ObjectId("660a82ed7c840f42b4a05533"),
    RollNo: 4,
    Age: 20,
    Cont: 4476,
    email: 'pani.de9@gmail.com'
  },
  {
    _id: ObjectId("660a82f47c840f42b4a05534"),
    RollNo: 10,
    Age: 23,
    Cont: 2276,
    email: 'rekha.de9@gmail.com'
  }
]
```

4. Write a query to update the Email-Id of a student with rollno 10.

```
{
  _id: ObjectId("660a83337c840f42b4a05535"),
  RollNo: 11,
  Age: 22,
  Name: 'ABC',
  Cont: 2276,
  email: 'rea.de9@gmail.com'
}
```

5. Replace the student name from “ABC” to “FEM” of rollno 11.

```
Atlas atlas-b6pfyk-shard-0 [primary] test> db.Student.update({RollNo:11,Name:"ABC"},{$set:{Name:"FEM"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

6. Export the created table into local file system

```
C:\Users\BMSCECSE>mongoexport mongodb+srv://antararc:Test1234@cluster0.mfnfeys.mongodb.net/myDB --collection=Student --out C:\Users\BMSCECSE\Downloads\output.json
2023-01-12T15:15:56.383+0530 connected to: mongodb+srv://[**REDACTED**]@cluster0.mfnfeys.mongodb.net/myDB
2023-01-12T15:15:56.497+0530 exported 7 records
```

7. Drop the table

```
Atlas atlas-b6pfyk-shard-0 [primary] test> db.Student.drop();
true
```

8. Import a given csv dataset from local file system into mongodb collection.

```
C:\Users\BMSCECSE>mongoimport mongodb+srv://antararc:Test1234@cluster0.mfnfeys.mongodb.net/myDB --collection=New_Student --type json --file C:\Users\BMSCECSE\Downloads\output.json
2023-01-12T15:17:35.523+0530 connected to: mongodb+srv://[**REDACTED**]@cluster0.mfnfeys.mongodb.net/myDB
2023-01-12T15:17:35.640+0530 7 document(s) imported successfully. 0 document(s) failed to import.
```

MongoDB- CRUD Operations Demonstration (Practice and Self Study)

Command with output using terminal:

```
test> use Employee
switched to db Employee
Employee> db
Employee
Employee> db.createCollection("Employee_Info")
{ ok: 1 }
Employee> show collections
Employee_Info
Employee> db.Employee_Info.insertMany([
...   {
...     Emp_Id: 121,
...     Emp_Name: "Amit",
...     Designation: "Software Engineer",
...     Date_of_Joining: ISODate("2019-06-15"),
...     Salary: 60000,
...     Dept_Name: "IT"
...   },
...   {
...     Emp_Id: 122,
...     Emp_Name: "Ravi",
...     Designation: "Senior Engineer",
...     Date_of_Joining: ISODate("2018-03-20"),
...     Salary: 75000,
...     Dept_Name: "R&D"
...   },
...   {
...     Emp_Id: 123,
...     Emp_Name: "Meera",
...     Designation: "Manager",
...     Date_of_Joining: ISODate("2016-01-10"),
...     Salary: 90000,
...     Dept_Name: "HR"
...   }
... ])
...
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69899e77e7bd2cd77163b112'),
    '1': ObjectId('69899e77e7bd2cd77163b113'),
    '2': ObjectId('69899e77e7bd2cd77163b114')
  }
}
```



```

mongosh mongodb://127.0.0.1
}
Employee> db.Employee_Info.find({ Emp_Id: 121 })
[
  {
    _id: ObjectId('69899e77e7bd2cd77163b112'),
    Emp_Id: 121,
    Emp_Name: 'Amit Kumar',
    Designation: 'Software Engineer',
    Date_of_Joining: ISODate('2019-06-15T00:00:00.000Z'),
    Salary: 60000,
    Dept_Name: 'Data Science'
  }
]
Employee> db.Employee_Info.find().sort({ Salary: -1 })
[
  {
    _id: ObjectId('69899e77e7bd2cd77163b114'),
    Emp_Id: 123,
    Emp_Name: 'Meera',
    Designation: 'Manager',
    Date_of_Joining: ISODate('2016-01-10T00:00:00.000Z'),
    Salary: 90000,
    Dept_Name: 'HR'
  },
  {
    _id: ObjectId('69899e77e7bd2cd77163b113'),
    Emp_Id: 122,
    Emp_Name: 'Ravi',
    Designation: 'Senior Engineer',
    Date_of_Joining: ISODate('2018-03-20T00:00:00.000Z'),
    Salary: 75000,
    Dept_Name: 'R&D'
  },
  {
    _id: ObjectId('69899e77e7bd2cd77163b112'),
    Emp_Id: 121,
    Emp_Name: 'Amit Kumar',
    Designation: 'Software Engineer',
    Date_of_Joining: ISODate('2019-06-15T00:00:00.000Z'),
    Salary: 60000,
    Dept_Name: 'Data Science'
  }
]
Employee> db.Employee_Info.find().sort({ Salary: 1 })
[
  {
    _id: ObjectId('69899e77e7bd2cd77163b112'),
    Emp_Id: 121,
    Emp_Name: 'Amit Kumar',
    Designation: 'Software Engineer',
    Date_of_Joining: ISODate('2019-06-15T00:00:00.000Z'),

```

```
mongosh mongodb://127.0.0.1:27027/employee_db?authSource=admin
> use employee_db
> db.Employee_Info.find()
{
  Designation: 'Software Engineer',
  Date_of_Joining: ISODate('2019-06-15T00:00:00.000Z'),
  Salary: 60000,
  Dept_Name: 'Data Science'
}
]
Employee> db.Employee_Info.find().sort({ Salary: -1 })
[
  {
    _id: ObjectId('69899e77e7bd2cd77163b114'),
    Emp_Id: 123,
    Emp_Name: 'Meera',
    Designation: 'Manager',
    Date_of_Joining: ISODate('2016-01-10T00:00:00.000Z'),
    Salary: 90000,
    Dept_Name: 'HR'
  },
  {
    _id: ObjectId('69899e77e7bd2cd77163b113'),
    Emp_Id: 122,
    Emp_Name: 'Ravi',
    Designation: 'Senior Engineer',
    Date_of_Joining: ISODate('2018-03-20T00:00:00.000Z'),
    Salary: 75000,
    Dept_Name: 'R&D'
  },
  {
    _id: ObjectId('69899e77e7bd2cd77163b112'),
    Emp_Id: 121,
    Emp_Name: 'Amit Kumar',
    Designation: 'Software Engineer',
    Date_of_Joining: ISODate('2019-06-15T00:00:00.000Z'),
    Salary: 60000,
    Dept_Name: 'Data Science'
  }
]
Employee> db.Employee_Info.find().sort({ Salary: 1 })
[
  {
    _id: ObjectId('69899e77e7bd2cd77163b112'),
    Emp_Id: 121,
    Emp_Name: 'Amit Kumar',
    Designation: 'Software Engineer',
    Date_of_Joining: ISODate('2019-06-15T00:00:00.000Z'),
    Salary: 60000,
    Dept_Name: 'Data Science'
  },
  {
    _id: ObjectId('69899e77e7bd2cd77163b113'),
    Emp_Id: 122,
    Emp_Name: 'Ravi',
    Designation: 'Senior Engineer',
    Date_of_Joining: ISODate('2018-03-20T00:00:00.000Z'),
    Salary: 75000,
    Dept_Name: 'R&D'
  },
  {
    _id: ObjectId('69899e77e7bd2cd77163b114'),
    Emp_Id: 123,
    Emp_Name: 'Meera',
    Designation: 'Manager',
    Date_of_Joining: ISODate('2016-01-10T00:00:00.000Z'),
    Salary: 90000,
    Dept_Name: 'HR'
  }
]
```

```

mongosh mongodb://127.0.0.1:27017/
> use Employee
Employee> db.Employee_Info.find({ Emp_Id: 121 })
[
  {
    Emp_Name: 'Ravi',
    Designation: 'Senior Engineer',
    Date_of_Joining: ISODate('2018-03-20T00:00:00.000Z'),
    Salary: 75000,
    Dept_Name: 'R&D'
  },
  {
    _id: ObjectId('69899e77e7bd2cd77163b114'),
    Emp_Id: 123,
    Emp_Name: 'Meera',
    Designation: 'Manager',
    Date_of_Joining: ISODate('2016-01-10T00:00:00.000Z'),
    Salary: 90000,
    Dept_Name: 'HR'
  }
]
Employee> db.Employee_Info.updateOne(
...   { Emp_Id: 121 },
...   { $set: { Projects: ["AI-Analytics", "Cloud-Migration", "BigData-Pipeline"] } }
... )
...
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Employee> db.Employee_Info.updateOne(
...   { Emp_Id: 121 },
...   { $addToSet: { Projects: "ML-Deployment" } }
... )
...
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Employee> db.Employee_Info.find({ Emp_Id: 121 })
[
  {
    _id: ObjectId('69899e77e7bd2cd77163b112'),
    Emp_Id: 121,
    Emp_Name: 'Amit Kumar',
    Designation: 'Software Engineer',
    Date_of_Joining: ISODate('2019-06-15T00:00:00.000Z'),
    Salary: 60000,
    Dept_Name: 'Data Science',
    Projects: [

```

```

mongosh mongodb://127.0.0.1:27017/
Employee> db.Employee_Info.find({ Emp_Id: 121 })
[
  {
    _id: ObjectId('69899e77e7bd2cd77163b112'),
    Emp_Id: 121,
    Emp_Name: 'Amit Kumar',
    Designation: 'Software Engineer',
    Date_of_Joining: ISODate('2019-06-15T00:00:00.000Z'),
    Salary: 60000,
    Dept_Name: 'Data Science',
    Projects: [
      'AI-Analytics',
      'Cloud-Migration',
      'BigData-Pipeline',
      'ML-Deployment'
    ]
  }
]
Employee> db.Employee_Info.insertOne({
...   Emp_Id: 130,
...   Emp_Name: "TemporaryUser",
...   Designation: "Intern",
...   Date_of_Joining: ISODate("2024-01-01"),
...   Salary: 15000,
...   Dept_Name: "Training",
...   createdAt: new Date()
... })
...
{
  acknowledged: true,
  insertedId: ObjectId('69899ef7e7bd2cd77163b115')
}
Employee> db.Employee_Info.createIndex(
...   { createdAt: 1 },
...   { expireAfterSeconds: 15 }
... )
...
createdAt_1
Employee> db.Employee_Info.find({ Emp_Id: 130 })
[
  {
    _id: ObjectId('69899ef7e7bd2cd77163b115'),
    Emp_Id: 130,
    Emp_Name: 'TemporaryUser',
    Designation: 'Intern',
    Date_of_Joining: ISODate('2024-01-01T00:00:00.000Z'),
    Salary: 15000,
    Dept_Name: 'Training',
    createdAt: ISODate('2026-02-09T08:46:47.854Z')
  }
]

```

PROGRAM – 2

Perform the following DB operations using Cassandra Questions:

- Create a keyspace by name Employee
- Create a column family by name
 - Employee-Info with attributes
 - Emp_Id Primary Key, Emp_Name,
 - Designation, Date_of_Joining,
 - Salary, Dept_Name
- Insert the values into the table in batch
- Update Employee name and Department of Emp-Id 121
- Sort the details of Employee records based on salary
- Alter the schema of the table Employee_Info to add a column Projects which stores a set of Projects done by the corresponding Employee.
- Update the altered table to add project names.
- Create a TTL of 15 seconds to display the values of Employees.

Command with output

Employee Database

1. Create a keyspace by name Employee

```
cqlsh:library> CREATE KEYSPACE Employee WITH REPLICATION = { 'class' : 'SimpleStrategy', 'replication_factor' : 1 };
cqlsh:library>
```

2. Create a column family by name Employee-Info with attributes Emp_Id Primary Key,

```
cqlsh:employee>
cqlsh:employee> CREATE TABLE Employee_Info (
...     Emp_Id int PRIMARY KEY,
...     Emp_Name text,
...     Designation text,
...     Date_of_Joining date,
...     Salary decimal,
...     Dept_Name text
... );
```

Emp_Name, Designation, Date_of_Joining, Salary, Dept_Name

3. Insert the values into the table in batch

```
cqlsh:employee> BEGIN BATCH
... INSERT INTO Employee_Info (Emp_Id, Emp_Name, Designation, Date_of_Joining, Salary, Dept_Name)
... VALUES (101, 'John Doe', 'Manager', '2023-01-01', 50000, 'HR');
... INSERT INTO Employee_Info (Emp_Id, Emp_Name, Designation, Date_of_Joining, Salary, Dept_Name)
... VALUES (121, 'Jane Smith', 'Developer', '2023-02-01', 60000, 'IT');
... APPLY BATCH;
cqlsh:employee> UPDATE Employee_Info SET Emp_Name = 'Jane Johnson', Dept_Name = 'Engineering' WHERE Emp_Id = 121;
cqlsh:employee> SELECT * FROM Employee_Info;
```

emp_id	date_of_joining	dept_name	designation	emp_name	salary
121	2023-02-01	Engineering	Developer	Jane Johnson	60000
101	2023-01-01	HR	Manager	John Doe	50000

(2 rows)

4. Update Employee name and Department of Emp-Id 121

5. Sort the details of Employee records based on salary


```

cqlsh:employee> paging off
Disabled Query paging.
cqlsh:employee> SELECT * FROM Employee_Info WHERE Emp_Id IN (121,101) ORDER BY Salary ALLOW FILTERING;

 emp_id | salary | date_of_joining | dept_name | designation | emp_name | projects
-----+-----+-----+-----+-----+-----+-----
    121 | 60000 | 2023-02-01 | IT | Developer | Jane Smith | {'ProjectC'}
    101 | 50000 | 2023-01-01 | HR | Manager | John Doe | {'ProjectA', 'ProjectB'}

(2 rows)

```

6. Alter the schema of the table Employee_Info to add a column Projects which stores a set of Projects done by the corresponding Employee.

7. Update the altered table to add project names.

8. Create a TTL of 15 seconds to display the values of Employees.

```

cqlsh:employee> UPDATE Employee_Info SET Projects = {'ProjectA', 'ProjectB'} WHERE Emp_Id = 101 and salary=50000;
cqlsh:employee> UPDATE Employee_Info SET Projects = {'ProjectC'} WHERE Emp_Id = 121 and salary=60000;
cqlsh:employee> select * from Employee_Info;

 emp_id | salary | date_of_joining | dept_name | designation | emp_name | projects
-----+-----+-----+-----+-----+-----+-----
    121 | 60000 | 2023-02-01 | IT | Developer | Jane Smith | {'ProjectC'}
    101 | 50000 | 2023-01-01 | HR | Manager | John Doe | {'ProjectA', 'ProjectB'}

(2 rows)

```

```

cqlsh:employee> INSERT INTO Employee_Info (Emp_Id, Emp_Name, Designation, Date_of_Joining, Salary, Dept_Name) VALUES (102, 'Jane Smith', 'Developer', '2022-06-03', 60000, 'IT') USING TTL 15;
cqlsh:employee> select ttl(Emp_Name) from Employee_Info where Emp_id=102;

 ttl(emp_name)
-----
          14

(1 rows)

```

PROGRAM – 3

Perform the following DB operations using Cassandra Questions:

- a) Create a keyspace by name Library
- b) Create a column family by name Library-Info with attributes
 - Stud_Id Primary Key,
 - Counter_value of type Counter,
 - Stud_Name, Book-Name, Book-Id,
 - Date_of_issue
- c) Insert the values into the table in batch
- d) Display the details of the table created and increase the value of the counter
- e) Write a query to show that a student with id 112 has taken a book “BDA” 2 times.
- f) Export the created column to a csv file
- g) Import a given csv dataset from local file system into Cassandra column family

Command with output

Use HELP for help.

```
cqlsh> DESCRIBE KEYSPACES;
```

```
system      system_distributed  system_traces  system_virtual_schema
system_auth  system_schema      system_views
```

```
cqlsh> CREATE KEYSPACE Library
```

```
... WITH replication = {'class': 'SimpleStrategy', 'replication_factor': 1};
```

```
cqlsh> USE Library;
```

```
cqlsh:library> CREATE TABLE Library_Info (
```

```
...     Stud_Id int,
...     Stud_Name text,
...     Book_Name text,
...     Book_Id int,
...     Date_of_issue date,
...     PRIMARY KEY (Stud_Id, Book_Id)
... );
```

```
cqlsh:library> CREATE TABLE Book_Count (
```

```
...     Stud_Id int,
...     Book_Name text,
...     Counter_value counter,
...     PRIMARY KEY (Stud_Id, Book_Name)
... );
```

```
cqlsh:library> BEGIN BATCH
```

```
... INSERT INTO Library_Info (Stud_Id, Stud_Name, Book_Name, Book_Id, Date_of_issue)
```

```
... VALUES (112, 'Rahul', 'BDA', 1, '2026-02-24');
```

```
...
```

```
... INSERT INTO Library_Info (Stud_Id, Stud_Name, Book_Name, Book_Id, Date_of_issue)
```

```
... VALUES (113, 'Anita', 'ML', 2, '2026-02-24');
```

```
... APPLY BATCH;
```

```
cqlsh:library> UPDATE Book_Count
```

```
... SET Counter_value = Counter_value + 1
```

```
... WHERE Stud_Id = 112 AND Book_Name = 'BDA';
```



```

... );
cqlsh:library> CREATE TABLE Book_Count (
...     Stud_Id int,
...     Book_Name text,
...     Counter_value counter,
...     PRIMARY KEY (Stud_Id, Book_Name)
... );
cqlsh:library> BEGIN BATCH
... INSERT INTO Library_Info (Stud_Id, Stud_Name, Book_Name, Book_Id, Date_of_issue)
... VALUES (112, 'Rahul', 'BDA', 1, '2026-02-24');
...
... INSERT INTO Library_Info (Stud_Id, Stud_Name, Book_Name, Book_Id, Date_of_issue)
... VALUES (113, 'Anita', 'ML', 2, '2026-02-24');
... APPLY BATCH;
cqlsh:library> UPDATE Book_Count
... SET Counter_value = Counter_value + 1
... WHERE Stud_Id = 112 AND Book_Name = 'BDA';
cqlsh:library>
cqlsh:library> UPDATE Book_Count
... SET Counter_value = Counter_value + 1
... WHERE Stud_Id = 112 AND Book_Name = 'BDA';
cqlsh:library> SELECT * FROM Library_Info;

```

stud_id	book_id	book_name	date_of_issue	stud_name
113	2	ML	2026-02-24	Anita
112	1	BDA	2026-02-24	Rahul

(2 rows)

```
cqlsh:library> SELECT * FROM Book_Count;
```

stud_id	book_name	counter_value
112	BDA	2

```

112 | BDA | 2
(1 rows)
cqlsh:library> UPDATE Book_Count
... SET Counter_value = Counter_value + 1
... WHERE Stud_Id = 112 AND Book_Name = 'BDA';
cqlsh:library> SELECT Counter_value
... FROM Book_Count
... WHERE Stud_Id = 112 AND Book_Name = 'BDA';

```

counter_value
3

(1 rows)

```
cqlsh:library> exit
```

```

C:\Users\BPC\Downloads\apache-cassandra-4.1.10-bin\apache-cassandra-4.1.10\bin>docker exec -it my-cassandra cqlsh -e "COPY Library.Library_Info
TO '/tmp/library.csv' WITH HEADER = true;"
Using 7 child processes

```

```

Starting copy of library.library_info with columns [stud_id, book_id, book_name, date_of_issue, stud_name].
Processed: 2 rows; Rate:      7 rows/s; Avg. rate:      3 rows/s
2 rows exported to 1 files in 0.633 seconds.

```

```

C:\Users\BPC\Downloads\apache-cassandra-4.1.10-bin\apache-cassandra-4.1.10\bin>docker cp my-cassandra:/tmp/library.csv library.csv
Successfully copied 2.05kB to C:\Users\BPC\Downloads\apache-cassandra-4.1.10-bin\apache-cassandra-4.1.10\bin\library.csv

```

```

C:\Users\BPC\Downloads\apache-cassandra-4.1.10-bin\apache-cassandra-4.1.10\bin>docker cp library.csv my-cassandra:/tmp/library.csv
Successfully copied 2.05kB to my-cassandra:/tmp/library.csv

```

```

C:\Users\BPC\Downloads\apache-cassandra-4.1.10-bin\apache-cassandra-4.1.10\bin>docker exec -it my-cassandra cqlsh -e "COPY Library.Library_Info
FROM '/tmp/library.csv' WITH HEADER = true;"
Using 7 child processes

```

```

Starting copy of library.library_info with columns [stud_id, book_id, book_name, date_of_issue, stud_name].
Processed: 2 rows; Rate:      4 rows/s; Avg. rate:      5 rows/s
2 rows imported from 1 files in 0.379 seconds (0 skipped).

```

AutoSave Off library.csv Search

File Home Insert Draw Page Layout Formulas Data Review View Help

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POSSIBLE DATA LOSS Some features might be lost if you save this workbook in the comma-delimited (.csv) format. To preserve these features, save it in an Excel file format. Don't show again Save As...

D5

	A	B	C	D	E	F	G
1	stud_id	book_id	book_name	date_of_issue	stud_name		
2	113	2	ML	24/02/2026	Anita		
3	112	1	BDA	24/02/2026	Rahul		
4							
5							
6							
7							
8							
9							
10							

PROGRAM – 4

Execution of HDFS Commands for interaction with Hadoop Environment. (Minimum 10 commands to be executed)

Command with output

1. mkdir
2. ls

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ start-all.sh
WARNING: Attempting to start all Apache Hadoop daemons as hadoop in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [bmscecse-HP-Elite-Tower-800-G9-Desktop-PC]
Starting resourcemanager
Starting nodemanagers
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -mkdir /bda_hadoop
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -ls /
Found 1 items
drwxr-xr-x  - hadoop supergroup          0 2024-05-13 14:37 /bda_hadoop
```

3. put

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -put /home/hadoop/Desktop/bda_local.txt /bda_hadoop/file.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -ls /bda_hadoop
Found 1 items
-rw-r--r--  1 hadoop supergroup          9 2024-05-13 14:42 /bda_hadoop/file.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -cat /bda_hadoop/file.txt
Hello!!!
```

4. copyFromLocal

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -copyFromLocal /home/hadoop/Desktop/bda_local.txt /bda_hadoop/file_cp_local.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -cat /bda_hadoop/file_cp_local.txt
Hello!!!
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$
```

5. get

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -get /bda_hadoop/file.txt /home/hadoop/Desktop/downloaded_file.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -getmerge /bda_hadoop/file.txt /bda_hadoop/file_cp_local.txt /home/hadoop/Desktop/downloaded_file.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -getfacl /bda_hadoop/
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -getfacl /bda_hadoop/
# file: /bda_hadoop
# owner: hadoop
# group: supergroup
user::rwx
group::r-x
other::r-x
```

6. copyToLocal

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -copyToLocal /bda_hadoop/file.txt /home/hadoop/Desktop
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -mv /bda_hadoop /abc
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -ls /abc
Found 2 items
-rw-r--r-- 1 hadoop supergroup          9 2024-05-13 14:42 /abc/file.txt
-rw-r--r-- 1 hadoop supergroup          9 2024-05-13 14:52 /abc/file_cp_local.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -cp /hello/ /hadoop_lab
cp: '/hello/': No such file or directory
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$
```

7. cat

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -cat /bda_hadoop/file_cp_local.txt
Hello!!!
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$
```

8. mv

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -mv /bda_hadoop /abc
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -ls /abc
Found 2 items
-rw-r--r-- 1 hadoop supergroup          9 2024-05-13 14:42 /abc/file.txt
-rw-r--r-- 1 hadoop supergroup          9 2024-05-13 14:52 /abc/file_cp_local.txt
```

9. cp

```
-rw-r--r-- 1 hadoop supergroup          9 2024-05-13 14:52 /abc/file_cp_local.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -cp /hello/ /hadoop_lab
cp: '/hello/': No such file or directory
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$
```

PROGRAM – 5

Implement Wordcount program on Hadoop framework

Driver Code

```
import java.io.IOException;
import org.apache.hadoop.conf.Configured;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.FileInputFormat;
import org.apache.hadoop.mapred.FileOutputFormat;
import org.apache.hadoop.mapred.JobClient;
import org.apache.hadoop.mapred.JobConf;
import org.apache.hadoop.util.Tool;
import org.apache.hadoop.util.ToolRunner;

public class WCDriver extends Configured implements Tool {

    public int run(String[] args) throws IOException {
        if (args.length < 2) {
            System.out.println("Please give valid inputs");
            return -1;
        }

        JobConf conf = new JobConf(WCDriver.class);
        conf.setJobName("WordCount");

        FileInputFormat.setInputPaths(conf, new Path(args[0]));
        FileOutputFormat.setOutputPath(conf, new Path(args[1]));

        conf.setMapperClass(WCMapper.class);
        conf.setReducerClass(WCReducer.class);

        conf.setMapOutputKeyClass(Text.class);
        conf.setMapOutputValueClass(IntWritable.class);

        conf.setOutputKeyClass(Text.class);
        conf.setOutputValueClass(IntWritable.class);

        JobClient.runJob(conf);
        return 0;
    }

    // Main Method
    public static void main(String[] args) throws Exception {
        int exitCode = ToolRunner.run(new WCDriver(), args);
        System.out.println("Job Exit Code: " + exitCode);
    }
}
```

Mapper Code

```
import java.io.IOException;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.MapReduceBase;
import org.apache.hadoop.mapred.Mapper;
import org.apache.hadoop.mapred.OutputCollector;
import org.apache.hadoop.mapred.Reporter;
```

```

public class WCMapper extends MapReduceBase implements Mapper<LongWritable, Text, Text, IntWritable> {

    // Map function
    public void map(LongWritable key, Text value, OutputCollector<Text, IntWritable> output, Reporter reporter)
        throws IOException {
        String line = value.toString();

        // Splitting the line on whitespace
        for (String word : line.split("\\s+")) {
            if (word.length() > 0) {
                output.collect(new Text(word), new IntWritable(1));
            }
        }
    }
}

```

Reducer Code

```

import
java.io.IOException; import
java.util.Iterator;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.MapReduceBase;
import org.apache.hadoop.mapred.OutputCollector;
import org.apache.hadoop.mapred.Reducer;
import org.apache.hadoop.mapred.Reporter;

public class WCReducer extends MapReduceBase implements Reducer<Text, IntWritable, Text, IntWritable> {

    // Reduce function
    public void reduce(Text key, Iterator<IntWritable> values,
        OutputCollector<Text, IntWritable> output,
        Reporter reporter) throws IOException {
        int count = 0;

        // Counting the frequency of each word
        while (values.hasNext()) {
            count += values.next().get();
        }

        output.collect(key, new IntWritable(count));
    }
}

```



```

hadoop@bmsccse-HP-Elite-Tower-600-G9-Desktop-PC:~$ mkdir WordCount
cd WordCount
hadoop@bmsccse-HP-Elite-Tower-600-G9-Desktop-PC:~/WordCount$ nano WCMapper.java
hadoop@bmsccse-HP-Elite-Tower-600-G9-Desktop-PC:~/WordCount$ nano WCReducer.java
hadoop@bmsccse-HP-Elite-Tower-600-G9-Desktop-PC:~/WordCount$ nano WCDriver.java
hadoop@bmsccse-HP-Elite-Tower-600-G9-Desktop-PC:~/WordCount$ export HADOOP_CLASSPATH=$(hadoop classpath)
hadoop@bmsccse-HP-Elite-Tower-600-G9-Desktop-PC:~/WordCount$ javac -classpath $HADOOP_CLASSPATH -d . WCMapper.java WCReducer.java WCDriver.java
hadoop@bmsccse-HP-Elite-Tower-600-G9-Desktop-PC:~/WordCount$ jar -cvf wordcount.jar *.class
added manifest
adding: WCDriver.class(in = 1397) (out= 689)(deflated 50%)
adding: WCMapper.class(in = 1857) (out= 757)(deflated 59%)
adding: WCReducer.class(in = 1529) (out= 605)(deflated 60%)
hadoop@bmsccse-HP-Elite-Tower-600-G9-Desktop-PC:~/WordCount$ cd /usr/local/hadoop/sbin
start-all.sh
jps
bash: cd: /usr/local/hadoop/sbin: No such file or directory
WARNING: Attempting to start all Apache Hadoop daemons in hadoop in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [bmsccse-HP-Elite-Tower-600-G9-Desktop-PC]
Starting resourcemanager
Starting nodemanagers
9152 Jps
8274 SecondaryNameNode
7977 DataNode
7802 NameNode
8570 ResourceManager
8732 NodeManager
hadoop@bmsccse-HP-Elite-Tower-600-G9-Desktop-PC:~/WordCount$ mkdir input
nano input/sample.txt
hadoop@bmsccse-HP-Elite-Tower-600-G9-Desktop-PC:~/WordCount$ hadoop fs -mkdir /inputbda
hadoop fs -put input/sample.txt /inputbda
hadoop@bmsccse-HP-Elite-Tower-600-G9-Desktop-PC:~/WordCount$ hadoop jar wordcount.jar WCDriver /inputbda/sample.txt /outputbda
2026-04-24 14:17:12,433 INFO Impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties
2026-04-24 14:17:12,470 INFO Impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
2026-04-24 14:17:12,470 INFO Impl.MetricsSystemImpl: JobTracker metrics system started
2026-04-24 14:17:12,477 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-04-24 14:17:12,579 INFO mapreduce.JobInputFormat: Total input files to process : 1
2026-04-24 14:17:12,607 INFO mapreduce.JobSubmitter: number of splits:1
2026-04-24 14:17:12,666 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_local1387788888_0001
2026-04-24 14:17:12,666 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-04-24 14:17:12,720 INFO mapreduce.Job: The url to track the job: http://localhost:8080/
2026-04-24 14:17:12,720 INFO mapreduce.Job: Running job: job_local1387788888_0001
2026-04-24 14:17:12,721 INFO mapred.LocalJobRunner: OutputCommitter set in config null
2026-04-24 14:17:12,722 INFO mapred.LocalJobRunner: OutputCommitter is org.apache.hadoop.mapred.FileOutputCommitter
2026-04-24 14:17:12,725 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-04-24 14:17:12,725 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2026-04-24 14:17:12,762 INFO mapred.LocalJobRunner: Waiting for map tasks

hadoop@bmsccse-HP-Elite-Tower-600-G9-Desktop-PC:~/WordCount$ hadoop fs -mkdir /inputbda
hadoop fs -put input/sample.txt /inputbda
hadoop@bmsccse-HP-Elite-Tower-600-G9-Desktop-PC:~/WordCount$ hadoop jar wordcount.jar WCDriver /inputbda/sample.txt /outputbda
2026-04-24 14:17:12,433 INFO Impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties
2026-04-24 14:17:12,470 INFO Impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
2026-04-24 14:17:12,470 INFO Impl.MetricsSystemImpl: JobTracker metrics system started
2026-04-24 14:17:12,533 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-04-24 14:17:12,579 INFO mapred.FileInputFormat: Total input files to process : 1
2026-04-24 14:17:12,607 INFO mapreduce.JobSubmitter: number of splits:1
2026-04-24 14:17:12,666 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_local1387788888_0001
2026-04-24 14:17:12,666 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-04-24 14:17:12,720 INFO mapreduce.Job: The url to track the job: http://localhost:8080/
2026-04-24 14:17:12,720 INFO mapreduce.Job: Running job: job_local1387788888_0001
2026-04-24 14:17:12,721 INFO mapred.LocalJobRunner: OutputCommitter set in config null
2026-04-24 14:17:12,722 INFO mapred.LocalJobRunner: OutputCommitter is org.apache.hadoop.mapred.FileOutputCommitter
2026-04-24 14:17:12,725 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-04-24 14:17:12,725 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2026-04-24 14:17:12,762 INFO mapred.LocalJobRunner: Waiting for map tasks
2026-04-24 14:17:12,764 INFO mapred.LocalJobRunner: Starting task: attempt_local1387788888_0001_m_000000_0
2026-04-24 14:17:12,775 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-04-24 14:17:12,775 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2026-04-24 14:17:12,780 INFO mapred.Task: Using ResourceCalculatorProcessTree : [ ]
2026-04-24 14:17:12,784 INFO mapred.MapTask: Processing split: hdfs://localhost:9000/inputbda/sample.txt:0+89
2026-04-24 14:17:12,793 INFO mapred.MapTask: numReduceTasks: 1
2026-04-24 14:17:12,821 INFO mapred.MapTask: (EQUATOR) 0 kvi 26214396(104857584)
2026-04-24 14:17:12,821 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2026-04-24 14:17:12,821 INFO mapred.MapTask: soft limit at 83886080
2026-04-24 14:17:12,821 INFO mapred.MapTask: bufstart = 0; bufvoid = 104857600
2026-04-24 14:17:12,821 INFO mapred.MapTask: kvstart = 26214396; length = 6553600
2026-04-24 14:17:12,823 INFO mapred.MapTask: Map output collector class = org.apache.hadoop.mapred.MapTask$MapOutputBuffer
2026-04-24 14:17:12,884 INFO mapred.LocalJobRunner:
2026-04-24 14:17:12,884 INFO mapred.MapTask: Starting flush of map output
2026-04-24 14:17:12,884 INFO mapred.MapTask: Spilling map output
2026-04-24 14:17:12,884 INFO mapred.MapTask: bufstart = 0; bufend = 169; bufvoid = 104857600
2026-04-24 14:17:12,884 INFO mapred.MapTask: kvstart = 26214396(104857584); kvoid = 26214320(104857280); length = 77/6553600
2026-04-24 14:17:12,887 INFO mapred.MapTask: Finished spill 0
2026-04-24 14:17:12,892 INFO mapred.Task: Task:attempt_local1387788888_0001_m_000000_0 is done. And is in the process of committing
2026-04-24 14:17:12,894 INFO mapred.LocalJobRunner: hdfs://localhost:9000/inputbda/sample.txt:0+89
2026-04-24 14:17:12,894 INFO mapred.Task: Task 'attempt_local1387788888_0001_m_000000_0' done.
2026-04-24 14:17:12,897 INFO mapred.Task: Final Counters for attempt_local1387788888_0001_m_000000_0: Counters: 23
File System Counters
  FILE: Number of bytes read=2920
  FILE: Number of bytes written=644387
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=89
  HDFS: Number of bytes written=0
  HDFS: Number of read operations=5
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=1
  HDFS: Number of bytes read erasure-coded=0
Map-Reduce Framework
  Map input records=5

```



```

Map input records=5
Map output records=20
Map output bytes=169
Map output materialized bytes=215
Input split bytes=93
Combine input records=0
Spilled Records=20
Failed Shuffles=0
Merged Map outputs=0
GC time elapsed (ms)=0
Total committed heap usage (bytes)=526385152
File Input Format Counters
  Bytes Read=89
2026-04-24 14:17:12,897 INFO mapred.LocalJobRunner: Finishing task: attempt_local1387788888_0001_r_000000_0
2026-04-24 14:17:12,897 INFO mapred.LocalJobRunner: map task executor complete.
2026-04-24 14:17:12,899 INFO mapred.LocalJobRunner: Waiting for reduce tasks
2026-04-24 14:17:12,899 INFO mapred.LocalJobRunner: Starting task: attempt_local1387788888_0001_r_000000_0
2026-04-24 14:17:12,902 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-04-24 14:17:12,902 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup_temporary folders under output directory:false, ignore cleanup failures: false
2026-04-24 14:17:12,902 INFO mapred.Task: Using ResourceCalculatorProcessFree: [ ]
2026-04-24 14:17:12,904 INFO mapred.ReduceTask: Using ShuffleConsumerPlugin: org.apache.hadoop.mapreduce.task.reduce.Shuffle@53ac8b1c
2026-04-24 14:17:12,904 WARN impl.MetricsSystemImpl: JobTracker metrics system already initialized!
2026-04-24 14:17:12,911 INFO reduce.MergeManagerImpl: MergerManager: memoryLimit=5830921216, maxSingleShuffleLimit=1457730304, mergeThreshold=3848408064, ioSortFactor=10, memToMemMergeOutputsThreshold=0
2026-04-24 14:17:12,912 INFO reduce.EventFetcher: attempt_local1387788888_0001_r_000000_0 Thread started: EventFetcher for fetching Map Completion Events
2026-04-24 14:17:12,925 INFO reduce.LocalFetcher: localFetcher#1 about to shuffle output of map attempt_local1387788888_0001_m_000000_0 decomp: 211 len: 215 to MEMORY
2026-04-24 14:17:12,926 INFO reduce.InMemoryMapOutput: Read 211 bytes from map-output for attempt_local1387788888_0001_m_000000_0
2026-04-24 14:17:12,927 INFO reduce.MergeManagerImpl: closeInMemoryFile -> map-output of size: 211, inMemoryMapOutputs.size() -> 1, commitMemory -> 0, usedMemory -> 211
2026-04-24 14:17:12,927 INFO reduce.EventFetcher: EventFetcher is interrupted.. Returning
2026-04-24 14:17:12,928 INFO mapred.LocalJobRunner: 1 / 1 copied.
2026-04-24 14:17:12,928 INFO reduce.MergeManagerImpl: finalMerge called with 1 in-memory map-outputs and 0 on-disk map-outputs
2026-04-24 14:17:12,930 INFO mapred.Merger: Merging 1 sorted segments
2026-04-24 14:17:12,930 INFO mapred.Merger: Down to the last merge-pass, with 1 segments left of total size: 205 bytes
2026-04-24 14:17:12,931 INFO reduce.MergeManagerImpl: Merged 1 segments, 211 bytes to disk to satisfy reduce memory limit
2026-04-24 14:17:12,931 INFO reduce.MergeManagerImpl: Merging 1 files, 215 bytes from disk
2026-04-24 14:17:12,931 INFO reduce.MergeManagerImpl: Merging 0 segments, 0 bytes from memory into reduce
2026-04-24 14:17:12,931 INFO mapred.Merger: Merging 1 sorted segments
2026-04-24 14:17:12,931 INFO mapred.Merger: Down to the last merge-pass, with 1 segments left of total size: 205 bytes
2026-04-24 14:17:12,931 INFO mapred.LocalJobRunner: 1 / 1 copied.
2026-04-24 14:17:12,991 INFO mapred.Task: Task:attempt_local1387788888_0001_r_000000_0 is done. And is in the process of committing
2026-04-24 14:17:12,992 INFO mapred.LocalJobRunner: 1 / 1 copied.
2026-04-24 14:17:12,992 INFO mapred.Task: Task:attempt_local1387788888_0001_r_000000_0 is allowed to commit now
2026-04-24 14:17:13,007 INFO output.FileOutputCommitter: Saved output of task 'attempt_local1387788888_0001_r_000000_0' to hdfs://localhost:9000/outputbda
2026-04-24 14:17:13,008 INFO mapred.LocalJobRunner: reduce > reduce
2026-04-24 14:17:13,008 INFO mapred.Task: Task 'attempt_local1387788888_0001_r_000000_0' done.
2026-04-24 14:17:13,008 INFO mapred.Task: Final Counters for attempt_local1387788888_0001_r_000000_0: Counters: 30
File System Counters
  FILE: Number of bytes read=3382
  FILE: Number of bytes written=644602
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=89
  HDFS: Number of bytes written=69
  HDFS: Number of read operations=10
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=0
2026-04-24 14:17:12,992 INFO mapred.LocalJobRunner: 1 / 1 copied.
2026-04-24 14:17:12,992 INFO mapred.Task: Task:attempt_local1387788888_0001_r_000000_0 is allowed to commit now
2026-04-24 14:17:13,007 INFO output.FileOutputCommitter: Saved output of task 'attempt_local1387788888_0001_r_000000_0' to hdfs://localhost:9000/outputbda
2026-04-24 14:17:13,008 INFO mapred.LocalJobRunner: reduce > reduce
2026-04-24 14:17:13,008 INFO mapred.Task: Task 'attempt_local1387788888_0001_r_000000_0' done.
2026-04-24 14:17:13,008 INFO mapred.Task: Final Counters for attempt_local1387788888_0001_r_000000_0: Counters: 30
File System Counters
  FILE: Number of bytes read=3382
  FILE: Number of bytes written=644602
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=89
  HDFS: Number of bytes written=69
  HDFS: Number of read operations=10
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=3
  HDFS: Number of bytes read erasure-coded=0
Ubuntu Software
uce Framework
  Combine input records=0
  Combine output records=0
  Reduce input groups=10
  Reduce shuffle bytes=215
  Reduce input records=20
  Reduce output records=10
  Spilled Records=20
  Shuffled Maps =1
  Failed Shuffles=0
  Merged Map outputs=1
  GC time elapsed (ms)=0
  Total committed heap usage (bytes)=526385152
Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0
File Output Format Counters
  Bytes Written=69
2026-04-24 14:17:13,008 INFO mapred.LocalJobRunner: Finishing task: attempt_local1387788888_0001_r_000000_0
2026-04-24 14:17:13,008 INFO mapred.LocalJobRunner: reduce task executor complete.
2026-04-24 14:17:13,723 INFO mapreduce.Job: Job job_local1387788888_0001 running in uber mode : false
2026-04-24 14:17:13,723 INFO mapreduce.Job: map 100% reduce 100%
2026-04-24 14:17:13,724 INFO mapreduce.Job: Job job_local1387788888_0001 completed successfully
2026-04-24 14:17:13,728 INFO mapreduce.Job: Counters: 36
File System Counters
  FILE: Number of bytes read=6302
  FILE: Number of bytes written=1288989
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=178
  HDFS: Number of bytes written=69
  HDFS: Number of read operations=15

```

```

FILE: Number of bytes written=1288989
FILE: Number of read operations=0
FILE: Number of large read operations=0
FILE: Number of write operations=0
HDFS: Number of bytes read=178
HDFS: Number of bytes written=69
HDFS: Number of read operations=15
HDFS: Number of large read operations=0
HDFS: Number of write operations=4
HDFS: Number of bytes read erasure-coded=0
Map-Reduce Framework
  Map input records=5
  Map output records=20
  Map output bytes=169
  Map output materialized bytes=215
  Input split bytes=93
  Combine input records=0
  Combine output records=0
  Reduce input groups=10
  Reduce shuffle bytes=215
  Reduce input records=20
  Reduce output records=10
  Spilled Records=40
  Shuffled Maps =1
  Failed Shuffles=0
  Merged Map outputs=1
  GC time elapsed (ms)=0
  Total committed heap usage (bytes)=1052770304
Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=89
File Output Format Counters
  Bytes Written=69
hadoop@bnsccscse-HP-Elite-Tower-600-G9-Desktop-PC: ~/wordCount$ hadoop fs -ls /outputbda
Found 2 items
-rw-r--r-- 1 hadoop supergroup      0 2026-04-24 14:17 /outputbda/_SUCCESS
-rw-r--r-- 1 hadoop supergroup    69 2026-04-24 14:17 /outputbda/part-00000
hadoop@bnsccscse-HP-Elite-Tower-600-G9-Desktop-PC: ~/wordCount$ hadoop fs -cat /outputbda/part-00000
are 1
brother 1
family 1
ht 1
how 5
is 4
job 1
sister 1
you 1
your 4
hadoop@bnsccscse-HP-Elite-Tower-600-G9-Desktop-PC: ~/wordCount$

```

PROGRAM – 6

Implement Weather program on Hadoop framework

Questions:

From the following link extract the weather data

<https://github.com/tomwhite/hadoopbook/tree/master/input/ncdc/all>

- a) Create a MapReduce program to find average temperature for each year from NCDC data set.
- b) find the mean max temperature for every month.

Driver Code

```
package temp;

import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;

import org.apache.hadoop.mapreduce.Job;

import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class AverageDriver {

    public static void main(String[] args) throws Exception {

        if (args.length != 2) {
            System.err.println("Please enter both input and output parameters.");
            System.exit(-1);
        }

        // Creating a configuration and job instance
        Configuration conf = new Configuration();
        Job job = Job.getInstance(conf, "Average Calculation");

        job.setJarByClass(AverageDriver.class);

        // Input and output paths
        FileInputFormat.addInputPath(job, new Path(args[0]));
        FileOutputFormat.setOutputPath(job, new Path(args[1]));

        // Setting mapper and reducer classes
        job.setMapperClass(AverageMapper.class);
        job.setReducerClass(AverageReducer.class);

        // Output key and value types
        job.setOutputKeyClass(Text.class);
        job.setOutputValueClass(IntWritable.class);

        // Submitting the job and waiting for it to complete
        System.exit(job.waitForCompletion(true) ? 0 : 1);
    }
}
```

Mapper Code

```
package temp;

import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class AverageMapper extends Mapper<LongWritable, Text, Text, IntWritable> {

    public static final int MISSING = 9999;

    @Override
    public void map(LongWritable key, Text value, Context context)
        throws IOException, InterruptedException {

        String line = value.toString();

        // Extract year from fixed position
        String year = line.substring(15, 19);
        int temperature;

        // Determine if there's a '+' sign
        if (line.charAt(87) == '+') {
            temperature = Integer.parseInt(line.substring(88, 92));
        } else {
            temperature = Integer.parseInt(line.substring(87, 92));
        }

        // Quality check character
        String quality = line.substring(92, 93);

        // Only emit if data is valid
        if (temperature != MISSING && quality.matches("[01459]")) {
            context.write(new Text(year), new IntWritable(temperature));
        }
    }
}
```

Reducer Code

```
package temp;

import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class AverageReducer extends Reducer<Text, IntWritable, Text, IntWritable> {

    @Override
    public void reduce(Text key, Iterable<IntWritable> values,
        Context context) throws IOException, InterruptedException {

        int sumTemp = 0;
        int count = 0;

        for (IntWritable value : values) {
            sumTemp += value.get();
            count++;
        }

        if (count > 0) {
            int average = sumTemp / count;
            context.write(key, new IntWritable(average));
        }
    }
}
```

```

find: '/home/lost-found': Permission denied
/home/hadoop/1901
hadoop@bmscsece-HP-Ellite-Tower-600-G9-Desktop-PC: ~/Desktop
hadoop@bmscsece-HP-Ellite-Tower-600-G9-Desktop-PC: ~/Desktop$ cd
hadoop@bmscsece-HP-Ellite-Tower-600-G9-Desktop-PC: $ ls
1901                                map.py                               public
1902                                map.py                               red1.py
Desktop                             MDriver.java                        red.py
Documents                           'MDriver.java'                     snap
Downloads                           Mapper.java                         templates
eclipse-workspace                   MReducer.java                      videos
hadoop                               mvn                                 wc.jar
hadoop-3.3.6.tar.gz                 'pl.class'                          wordcount
hadoopdata                           pl.class                           wordcount
lnp.txt                             'icurus'                           wordcount.jar
input.txt                           pl.scala                            word.jar
t.txt                               'pl.class'                          wordcount

hadoop@bmscsece-HP-Ellite-Tower-600-G9-Desktop-PC: $ hdfs dfs -put 1901 /weather
hadoop@bmscsece-HP-Ellite-Tower-600-G9-Desktop-PC: $ hdfs dfs -put 1902 /weather
hadoop@bmscsece-HP-Ellite-Tower-600-G9-Desktop-PC: $ hdfs dfs -ls /weather
Found 2 items
-rw-r--r-- 1 hadoop supergroup      888190 2026-03-13 15:26 /weather/1901
-rw-r--r-- 1 hadoop supergroup      888978 2026-03-13 15:26 /weather/1902
hadoop@bmscsece-HP-Ellite-Tower-600-G9-Desktop-PC: $ cd -
hadoop@bmscsece-HP-Ellite-Tower-600-G9-Desktop-PC: $ ls
1901                                input.txt                           'pl.class'                          wc.jar
1902                                t.txt                              pl.class                           WeatherAnalysis
Desktop                             map.py                             pl.scala                           wordcount
Documents                           MDriver.java                       Public                             wordcount
Downloads                           'MDriver.java'                     red1.py                           wordcount
hadoop                               Mapper.java                        red.py                            wordcount.jar
hadoop-3.3.6.tar.gz                 MReducer.java                      snap                               word.jar
hadoopdata                           mvn                                 Templates                         videos
lnp.txt                             'pl.class'                          Videos
input.txt                           'pl.class'                          Videos
hadoop@bmscsece-HP-Ellite-Tower-600-G9-Desktop-PC: $ jps
7936 NameNode
8740 ResourceManager
10039 org.eclipse.equinox.launcher_1.7.100.v20251111-0406.jar
16696 Jps
8908 NodeManager
8413 SecondaryNameNode
8127 DataNode
hadoop@bmscsece-HP-Ellite-Tower-600-G9-Desktop-PC: $ hdfs dfs -ls /weather
Found 2 items
-rw-r--r-- 1 hadoop supergroup      888190 2026-03-13 15:26 /weather/1901
-rw-r--r-- 1 hadoop supergroup      888978 2026-03-13 15:26 /weather/1902
hadoop@bmscsece-HP-Ellite-Tower-600-G9-Desktop-PC: $ hadoop jar Weather.jar avgtemp.AverageDriver /weather /avgoutput
Exception in thread "main" java.lang.UnsupportedClassVersionError: avgtemp.AverageDriver has been compiled by a more recent version of the Java Runtime (class file version 65.0), this version of the Java
Runtime only recognizes class file versions up to 55.0
at java.base/java.lang.ClassLoader.defineClass1(Native Method)
at java.base/java.lang.ClassLoader.defineClass(ClassLoader.java:1022)
at java.base/java.security.SecureClassLoader.defineClass(SecureClassLoader.java:174)
at java.base/java.net.URLClassLoader.defineClass(URLClassLoader.java:555)
at java.base/java.net.URLClassLoader$1.run(URLClassLoader.java:458)

```

```

10039 org.eclipse.equinox.launcher_1.7.100.v20251111-0406.jar
16696 Jps
8908 NodeManager
8413 SecondaryNameNode
8127 DataNode
hadoop@bmscsece-HP-Ellite-Tower-600-G9-Desktop-PC: $ hdfs dfs -ls /weather
Found 2 items
-rw-r--r-- 1 hadoop supergroup      888190 2026-03-13 15:26 /weather/1901
-rw-r--r-- 1 hadoop supergroup      888978 2026-03-13 15:26 /weather/1902
hadoop@bmscsece-HP-Ellite-Tower-600-G9-Desktop-PC: $ hadoop jar Weather.jar avgtemp.AverageDriver /weather /avgoutput
Exception in thread "main" java.lang.UnsupportedClassVersionError: avgtemp.AverageDriver has been compiled by a more recent version of the Java Runtime (class file version 65.0), this version of the Java
Runtime only recognizes class file versions up to 55.0
at java.base/java.lang.ClassLoader.defineClass1(Native Method)
at java.base/java.lang.ClassLoader.defineClass(ClassLoader.java:1022)
at java.base/java.security.SecureClassLoader.defineClass(SecureClassLoader.java:174)
at java.base/java.net.URLClassLoader.defineClass(URLClassLoader.java:555)
at java.base/java.net.URLClassLoader$1.run(URLClassLoader.java:458)
at java.base/java.net.URLClassLoader$1.run(URLClassLoader.java:452)
at java.base/java.security.AccessController.doPrivileged(Native Method)
at java.base/java.net.URLClassLoader.findClass(URLClassLoader.java:451)
at java.base/java.lang.ClassLoader.loadClass(ClassLoader.java:594)
at java.base/java.lang.ClassLoader.loadClass(ClassLoader.java:527)
at java.base/java.lang.Class.forName0(Native Method)
at java.base/java.lang.Class.forName(Class.java:398)
at org.apache.hadoop.util.RunJar.run(RunJar.java:321)
at org.apache.hadoop.util.RunJar.main(RunJar.java:241)
hadoop@bmscsece-HP-Ellite-Tower-600-G9-Desktop-PC: $ hadoop jar Weather.jar avgtemp.AverageDriver /weather /avgoutput
2026-03-13 15:58:55.743 INFO client.DefaultHadoopClusterProvider: Connecting to ResourceManager at /p.s.b.0:8032
2026-03-13 15:58:55.936 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-03-13 15:58:56.117 INFO InputFileInputFormat: total input files to process : 2
2026-03-13 15:58:56.202 INFO mapreduce.JobSubmitter: number of splits:2
2026-03-13 15:58:56.319 INFO mapreduce.JobSubmitter: Submitting tokens for job: Job_177339556039_0001
2026-03-13 15:58:56.522 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-03-13 15:58:56.398 INFO conf.Configuration: resource-types.xml not found
2026-03-13 15:58:56.399 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-03-13 15:58:56.583 INFO impl.YarnClientImpl: Submitted application application_177339556039_0001
2026-03-13 15:58:56.522 INFO mapreduce.Job: The url to track the job: http://bmscsece-HP-Ellite-Tower-600-G9-Desktop-PC:8088/proxy/application_177339556039_0001/
2026-03-13 15:58:56.559 INFO mapreduce.Job: Job Job_177339556039_0001 running in uber mode : false
2026-03-13 15:59:00.568 INFO mapreduce.Job: map 0% reduce 0%
2026-03-13 15:59:03.586 INFO mapreduce.Job: map 100% reduce 0%
2026-03-13 15:59:07.598 INFO mapreduce.Job: map 100% reduce 100%
2026-03-13 15:59:08.687 INFO mapreduce.Job: Job Job_177339556039_0001 completed successfully
2026-03-13 15:59:08.652 INFO mapreduce.Job: Counters: 54
File System Counters
FILE: Number of bytes read=14425
FILE: Number of bytes written=1116570
FILE: Number of read operations=0
FILE: Number of write operations=0
HDFS: Number of bytes read=177360
HDFS: Number of bytes written=16
HDFS: Number of read operations=11
HDFS: Number of large read operations=0

```

```

HDFS: Number of large read operations=0
HDFS: Number of write operations=2
HDFS: Number of bytes read erasure-coded=0
Job Counters
Launched Reduce tasks=1
Total time spent by all maps in occupied slots (ms)=2234
Total time spent by all reduce in occupied slots (ms)=1123
Total time spent by all map tasks (ms)=2234
Total time spent by all reduce tasks (ms)=1123
Total mapreduce time taken by all map tasks (ms)=2234
Total mapreduce time taken by all reduce tasks (ms)=1123
Total mapreduce time taken by all map tasks (ms)=2234
Total mapreduce time taken by all reduce tasks (ms)=1123
Map-Reduce
Map Input records=13130
Map output records=13129
Map output bytes=118155
Map output materialized bytes=144431
Combine input records=0
Combine output records=0
Reduce input groups=2
Reduce shuffle bytes=144431
Reduce input records=13129
Reduce output records=2
Shuffled Map=2
Merged Map outputs=2
CPU time spent (ms)=2140
Physical memory (bytes) snapshot=107902704
Virtual memory (bytes) snapshot=107902704
Total mapreduce time taken by all map tasks (ms)=2234
Total mapreduce time taken by all reduce tasks (ms)=1123
Peak map virtual memory (bytes)=278757760
Peak reduce virtual memory (bytes)=278757760
Shuffle Errors
ERROR
IOError: java.io.IOException
HADOOP_JOB_ID=job_177339556039_0001
HADOOP_JOB_NAME=mapreduce
HADOOP_JOB_STATUS=RUNNING
File Input Format Counters
Bytes Read=177360
File Output Format Counters
Bytes Written=16
hadoop@bmscsece-HP-Ellite-Tower-600-G9-Desktop-PC: $ hdfs dfs -cat /avgoutput/part-r-000000
1901 40
1902 21
hadoop@bmscsece-HP-Ellite-Tower-600-G9-Desktop-PC: $

```

Driver Code

```
package meanmax;

import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;

import org.apache.hadoop.mapreduce.Job;

import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class MeanMaxDriver {

    public static void main(String[] args) throws Exception {

        if (args.length != 2) {
            System.err.println("Please enter both input and output parameters.");
            System.exit(-1);
        }

        Configuration conf = new Configuration();
        Job job = Job.getInstance(conf, "Mean and Max Temperature");

        job.setJarByClass(MeanMaxDriver.class);

        FileInputFormat.addInputPath(job, new Path(args[0]));
        FileOutputFormat.setOutputPath(job, new Path(args[1]));

        job.setMapperClass(MeanMaxMapper.class);
        job.setReducerClass(MeanMaxReducer.class);

        job.setOutputKeyClass(Text.class);
        job.setOutputValueClass(IntWritable.class);

        System.exit(job.waitForCompletion(true) ? 0 : 1);
    }
}
```

Mapper Code

```
package meanmax;

import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;

import org.apache.hadoop.mapreduce.Mapper;

public class MeanMaxMapper extends Mapper<LongWritable, Text, Text, IntWritable> {

    public static final int MISSING = 9999;

    @Override
    public void map(LongWritable key, Text value, Context context)
        throws IOException, InterruptedException {

        String line = value.toString();

        // Extract month from positions 19-20
        String month = line.substring(19, 21);
        int temperature;
```



```

// Extract temperature considering optional '+'
if (line.charAt(87) == '+') {
    temperature = Integer.parseInt(line.substring(88, 92));
} else {
    temperature = Integer.parseInt(line.substring(87, 92));
}

// Quality check
String quality = line.substring(92, 93);

if (temperature != MISSING && quality.matches("[01459]")) {
    context.write(new Text(month), new IntWritable(temperature));
}
}
}

```

Reducer Code

```

package meanmax;

import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class MeanMaxReducer extends Reducer<Text, IntWritable, Text, Text> {

    @Override
    public void reduce(Text key, Iterable<IntWritable> values,
        Context context) throws IOException, InterruptedException {

        int sumTemp = 0;
        int count = 0;
        int maxTemp = Integer.MIN_VALUE;

        for (IntWritable value : values) {
            int temp = value.get();
            sumTemp += temp;
            count++;

            if (temp > maxTemp) {
                maxTemp = temp;
            }
        }

        if (count > 0) {
            int avgTemp = sumTemp / count;
            String result = "mean=" + avgTemp + " max=" + maxTemp;
            context.write(key, new Text(result));
        }
    }
}

```

```

1901 46
1902 21
hadoop@bmsccse-HP-Ellite-Tower-600-G9-Desktop-PC: $ hdfs dfs -rm -r /meannaxoutput
rm: '/meannaxoutput': No such file or directory
hadoop@bmsccse-HP-Ellite-Tower-600-G9-Desktop-PC: $ hadoop jar Weather.jar meannax.MeanMaxDriver /weather /meannaxoutput
2026-03-13 16:05:30,143 INFO client.DefaultHARMFalloverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-03-13 16:05:30,453 INFO Input.FileInputFormat: Total input files to process : 2
2026-03-13 16:05:30,529 INFO mapreduce.JobSubmitter: number of splits:2
2026-03-13 16:05:30,612 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1773395565039_0002
2026-03-13 16:05:30,612 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-03-13 16:05:30,682 INFO conf.Configuration: resource-types.xml not found
2026-03-13 16:05:30,682 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'
2026-03-13 16:05:30,709 INFO Inpl.VarnClientImpl: Submitted application application_1773395565039_0002
2026-03-13 16:05:30,725 INFO mapreduce.Job: The url to track the job: http://bmsccse-HP-Ellite-Tower-600-G9-Desktop-PC:8088/proxy/application_1773395565039_0002/
2026-03-13 16:05:30,726 INFO mapreduce.Job: Running job: job_1773395565039_0002
2026-03-13 16:05:34,760 INFO mapreduce.Job: Job job_1773395565039_0002 running in uber mode : false
2026-03-13 16:05:34,761 INFO mapreduce.Job: map 0% reduce 0%
2026-03-13 16:05:37,786 INFO mapreduce.Job: map 100% reduce 0%
2026-03-13 16:05:41,800 INFO mapreduce.Job: map 100% reduce 100%
2026-03-13 16:05:41,806 INFO mapreduce.Job: Job job_1773395565039_0002 completed successfully
2026-03-13 16:05:41,847 INFO mapreduce.Job: Counters: 54
File System Counters
  FILE: Number of bytes read=118167
  FILE: Number of bytes written=1064069
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=1777366
  HDFS: Number of bytes written=72
  HDFS: Number of read operations=11
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=2
  HDFS: Number of bytes read erasure-coded=0
Job Counters
  Launched map tasks=2
  Launched reduce tasks=1
  Data-local map tasks=2
  Total time spent by all maps in occupied slots (ms)=2213
  Total time spent by all reduces in occupied slots (ms)=1047
  Total time spent by all map tasks (ms)=2213
  Total time spent by all reduce tasks (ms)=1047
  Total vcore-milliseconds taken by all map tasks=2213
  Total vcore-milliseconds taken by all reduce tasks=1047
  Total megabyte-milliseconds taken by all map tasks=2266112
  Total megabyte-milliseconds taken by all reduce tasks=1072128
Map-Reduce Framework
  Map input records=13130
  Map output records=13129
  Map output bytes=91903
  Map output materialized bytes=118173
  Input split bytes=198
  Combine input records=0
  Combine output records=0
  Combine input records=0
  Combine output records=0

```

```

Total time spent by all reduce tasks (ms)=1047
Total vcore-milliseconds taken by all map tasks=2213
Total vcore-milliseconds taken by all reduce tasks=1047
Total megabyte-milliseconds taken by all map tasks=2266112
Total megabyte-milliseconds taken by all reduce tasks=1072128
Map-Reduce Framework
  Map input records=13130
  Map output records=13129
  Map output bytes=91903
  Map output materialized bytes=118173
  Input split bytes=198
  Combine input records=0
  Combine output records=0
  Reduce input groups=12
  Reduce shuffle bytes=118173
  Reduce input records=13129
  Reduce output records=12
  Spilled Records=26258
  Shuffled Maps =2
  Failed Shuffles=0
  Merged Map outputs=2
  GC time elapsed (ms)=15
  CPU time spent (ms)=2030
  Physical memory (bytes) snapshot=1037684736
  Virtual memory (bytes) snapshot=8176963072
  Total committed heap usage (bytes)=157286400
  Peak Map Physical memory (bytes)=381546496
  Peak Map Virtual memory (bytes)=2792185856
  Peak Reduce Physical memory (bytes)=281702400
  Peak Reduce Virtual memory (bytes)=2799857664
Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=1777168
File Output Format Counters
  Bytes Written=72
hadoop@bmsccse-HP-Ellite-Tower-600-G9-Desktop-PC: $ hdfs dfs -cat /meannaxoutput/part-r-00000
01 4
02 1
03 6
04 34
05 89
06 143
07 182
08 172
09 123
10 73
11 21
12 3
hadoop@bmsccse-HP-Ellite-Tower-600-G9-Desktop-PC: $

```

PROGRAM – 7

For a given Text file, Create a Map Reduce program to sort the content in an alphabetic order listing only top 10 maximum occurrences of words.

Driver Code (TopNDriver.java)

```
package samples.topn;

import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;

import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class TopNDriver {

    public static void main(String[] args) throws Exception {
        if (args.length != 3) {
            System.err.println("Usage: TopNDriver <in> <temp-out> <final-out>");
            System.exit(2);
        }

        Configuration conf = new Configuration();

        // == Job 1: Word Count ==
        Job wcJob = Job.getInstance(conf, "word count");
        wcJob.setJarByClass(TopNDriver.class);
        wcJob.setMapperClass(WordCountMapper.class);
        wcJob.setCombinerClass(WordCountReducer.class);
        wcJob.setReducerClass(WordCountReducer.class);
        wcJob.setOutputKeyClass(Text.class);
        wcJob.setOutputValueClass(IntWritable.class);

        FileInputFormat.addInputPath(wcJob, new Path(args[0]));
        Path tempDir = new Path(args[1]);
        FileOutputFormat.setOutputPath(wcJob, tempDir);

        if (!wcJob.waitForCompletion(true)) {
            System.exit(1);
        }

        // == Job 2: Top N ==
        Job topJob = Job.getInstance(conf, "top 10 words");
        topJob.setJarByClass(TopNDriver.class);
        topJob.setMapperClass(TopNMapper.class);
        topJob.setReducerClass(TopNReducer.class);
        topJob.setMapOutputKeyClass(IntWritable.class);
        topJob.setMapOutputValueClass(Text.class);
        topJob.setOutputKeyClass(Text.class);
        topJob.setOutputValueClass(IntWritable.class);

        FileInputFormat.addInputPath(topJob, tempDir);
        FileOutputFormat.setOutputPath(topJob, new Path(args[2]));

        System.exit(topJob.waitForCompletion(true) ? 0 : 1);
    }
}
```

Mapper Code (WordCountMapper.java)

```
package samples.topn;

import java.io.IOException;
import java.util.StringTokenizer;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class WordCountMapper
    extends Mapper<Object, Text, Text, IntWritable> {

    private final static IntWritable ONE = new IntWritable(1);
    private Text word = new Text();
    // characters to normalize into spaces
    private String tokens = "[_\\$#<>\\^=\\[\\]\\|\\*\\/\\\\\\\\\\,;,:\\-\\.()?!\"'"]";

    @Override
    protected void map(Object key, Text value, Context context)
        throws IOException, InterruptedException {

        // clean & tokenize
        String clean = value.toString()
            .toLowerCase()
            .replaceAll(tokens, " ");
        StringTokenizer itr = new StringTokenizer(clean);
        while (itr.hasMoreTokens()) {
            word.set(itr.nextToken().trim());
            context.write(word, ONE);
        }
    }
}
```

Mapper Code (TopNMapper.java)

```
package samples.topn;

import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class TopNMapper
    extends Mapper<Object, Text, IntWritable, Text> {

    private IntWritable count = new IntWritable();
    private Text word = new Text();

    @Override
    protected void map(Object key, Text value, Context context)
        throws IOException, InterruptedException {

        // input line: word \t count
        String[] parts = value.toString().split("\\t");
        if (parts.length == 2) {
            word.set(parts[0]);
            count.set(Integer.parseInt(parts[1]));
            // emit count → word, so Hadoop sorts by count
            context.write(count, word);
        }
    }
}
```

Reducer Code (WordCountReducer.java)

```
package samples.topn;

import java.io.IOException;
```

```

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class WordCountReducer
    extends Reducer<Text, IntWritable, Text, IntWritable> {

    @Override
    protected void reduce(Text key, Iterable<IntWritable> values, Context context)
        throws IOException, InterruptedException {

        int sum = 0;
        for (IntWritable val : values) {
            sum += val.get();
        }
        context.write(key, new IntWritable(sum));
    }
}

```

Reducer Code (TopNReducer.java)

```

package samples.topn;

import java.io.IOException;
import java.util.ArrayList;
import java.util.Collections;
import java.util.List;
import java.util.Map;
import java.util.TreeMap;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class TopNReducer
    extends Reducer<IntWritable, Text, Text, IntWritable> {

    // TreeMap with descending order of keys (counts)
    private TreeMap<Integer, List<String>> countMap =
        new TreeMap<>(Collections.reverseOrder());

    @Override
    protected void reduce(IntWritable key, Iterable<Text> values, Context context)
        throws IOException, InterruptedException {

        int cnt = key.get();
        List<String> words = countMap.getOrDefault(cnt, new ArrayList<>());
        for (Text w : values) {
            words.add(w.toString());
        }
        countMap.put(cnt, words);
    }

    @Override
    protected void cleanup(Context context)
        throws IOException, InterruptedException {

        // collect top 10 word→count pairs
        List<WordCount> topList = new ArrayList<>();
        int seen = 0;
        for (Map.Entry<Integer, List<String>> entry : countMap.entrySet()) {
            int cnt = entry.getKey();
            for (String w : entry.getValue()) {
                topList.add(new WordCount(w, cnt));
                seen++;
                if (seen == 10) break;
            }
            if (seen == 10) break;
        }
    }
}

```

```

    }

    // sort these 10 entries alphabetically by word
    Collections.sort(topList, (a, b) -> a.word.compareTo(b.word));

    // emit final top 10 in alphabetical order
    for (WordCount wc : topList) {
        context.write(new Text(wc.word), new IntWritable(wc.count));
    }
}

// helper class
private static class WordCount {
    String word;
    int count;
    WordCount(String w, int c) { word = w; count = c; }
}
}

```

GNU nano 6.2 /home/hadoop/sample.txt

```

hi how are you
how is your job
how is your family
how is your brother
how is your sister

```

Read 5 lines

^G Help	^O Write Out	^W Where Is	^K Cut	^T Execute	^C Location
^X Exit	^R Read File	^_ Replace	^U Paste	^J Justify	^/ Go To Line


```

hadop@bmscscce-HP-Elite-Tower-600-G9-Desktop-PC:~/hadoop$ ./start-all.sh
WARNING: Attempting to start all Apache Hadoop daemons as hadoop in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [bmscscce-HP-Elite-Tower-600-G9-Desktop-PC]
Starting resourcemanager
Starting nodemanagers
hadop@bmscscce-HP-Elite-Tower-600-G9-Desktop-PC:~/hadoop$ jps
31057 Jps
30787 NodeManager
30149 SecondaryNameNode
29706 NameNode
29902 DataNode
30431 ResourceManager
hadop@bmscscce-HP-Elite-Tower-600-G9-Desktop-PC:~/hadoop$ cd ~/topn
hadop@bmscscce-HP-Elite-Tower-600-G9-Desktop-PC:~/topn$ javac -classpath 'hadoop classpath' -d . TopN.java TopMapper.java TopReducer.java TopCombiner.java utils/MiscUtils.java
Note: utils/MiscUtils.java uses unchecked or unsafe operations.
Note: Recompile with -Xlint:unchecked for details.
hadop@bmscscce-HP-Elite-Tower-600-G9-Desktop-PC:~/topn$ jar -cvf TopN.jar *.class utils/*.class
added manifest
adding: TopN.class(in = 1388) (out= 764)(deflated 44%)
adding: TopCombiner.class(in = 1587) (out= 662)(deflated 58%)
adding: TopMapper.class(in = 1953) (out= 883)(deflated 54%)
adding: TopReducer.class(in = 2532) (out= 1003)(deflated 60%)
adding: utils/MiscUtils$1.class(in = 944) (out= 494)(deflated 47%)
adding: utils/MiscUtils.class(in = 1231) (out= 693)(deflated 43%)
hadop@bmscscce-HP-Elite-Tower-600-G9-Desktop-PC:~/topn$ nano ~/sample.txt
hadop@bmscscce-HP-Elite-Tower-600-G9-Desktop-PC:~/topn$ hadoop fs -mkdir /topninput
mkdir: '/topninput': File exists
hadop@bmscscce-HP-Elite-Tower-600-G9-Desktop-PC:~/topn$ hadoop fs -put ~/sample.txt /topninput
put: '/topninput/sample.txt': File exists
hadop@bmscscce-HP-Elite-Tower-600-G9-Desktop-PC:~/topn$ hadoop fs -ls /topninput
Found 1 items
-rw-r--r-- 1 hadoop supergroup 89 2026-04-24 14:58 /topninput/sample.txt
hadop@bmscscce-HP-Elite-Tower-600-G9-Desktop-PC:~/topn$ hadoop jar TopN.jar TopN /topninput/sample.txt /topnoutput
2026-04-24 15:04:22,185 INFO client.DefaultHARMFalloverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
Exception in thread "main" org.apache.hadoop.mapred.FileAlreadyExistsException: Output directory hdfs://localhost:9000/topnoutput already exists
    at org.apache.hadoop.mapreduce.lib.output.FileOutputFormat.checkOutputSpecs(FileOutputFormat.java:164)
    at org.apache.hadoop.mapreduce.JobSubmitter.checkSpecs(JobSubmitter.java:277)
    at org.apache.hadoop.mapreduce.JobSubmitter.submitJobInternal(JobSubmitter.java:143)
    at org.apache.hadoop.mapreduce.Job$11.run(Job.java:1678)
    at org.apache.hadoop.mapreduce.Job$11.run(Job.java:1675)
    at java.base/java.security.AccessController.doPrivileged(Native Method)
    at java.base/javax.security.auth.Subject.doAs(Subject.java:423)
    at org.apache.hadoop.security.UserGroupInformation.doAs(UserGroupInformation.java:1899)
    at org.apache.hadoop.mapreduce.Job.submit(Job.java:1675)
    at org.apache.hadoop.mapreduce.Job.waitForCompletion(Job.java:1696)
    at TopN.main(TopN.java:27)
    at java.base/jdk.internal.reflect.NativeMethodAccessorImpl.invoke0(Native Method)
    at java.base/jdk.internal.reflect.NativeMethodAccessorImpl.invoke(Native Method)
    at java.base/jdk.internal.reflect.DelegatingMethodAccessorImpl.invoke(DelegatingMethodAccessorImpl.java:43)
    at java.base/java.lang.reflect.Method.invoke(Method.java:566)
    at org.apache.hadoop.util.RunJar.run(RunJar.java:328)
    at org.apache.hadoop.util.RunJar.main(RunJar.java:241)
hadop@bmscscce-HP-Elite-Tower-600-G9-Desktop-PC:~/topn$ hadoop fs -cat /topnoutput/part-r-00000
how 5
your 4
is 4
brother 1
are 1
hi 1
sister 1
family 1
you 1
job 1
hadop@bmscscce-HP-Elite-Tower-600-G9-Desktop-PC:~/topn$

```

PROGRAM – 8

Write a Scala program to print numbers from 1 to 100 using for loop.

```
object ExampleForLoop1 {  
  def main(args: Array[String]) { for(counter  
    <- 1 to 100) {  
    print(counter + " ")  
  }  
  println()  
}
```

Output:

```
hadoop@bmscscse-HP-Elite-Tower-600-G9-Desktop-PC: $ scala -version  
Scala code runner version 2.12.18 -- Copyright 2002-2023, LAMP/EPFL and Lightbend, Inc.  
hadoop@bmscscse-HP-Elite-Tower-600-G9-Desktop-PC: $ nano PrintNumbers.scala  
hadoop@bmscscse-HP-Elite-Tower-600-G9-Desktop-PC: $ scala PrintNumbers.scala  
1  
2  
3  
4  
5  
6  
7  
8  
9  
10  
11  
12  
13  
14  
15  
16  
17  
18  
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32  
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36  
37  
38  
39  
40  
41  
42  
43  
44  
45  
46  
47
```

```
48  
49  
50  
51  
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53  
54  
55  
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58  
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61  
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92  
93  
94  
95  
96  
97  
98  
99  
100  
hadoop@bmscscse-HP-Elite-Tower-600-G9-Desktop-PC: $
```

PROGRAM – 9

Using RDD and FlatMap count how many times each word appears in a file and write out a list of words whose count is strictly greater than 4 using Spark.

```
scala> val textFile = sc.textFile("D:\\Downloads\\spark-4.1.1-bin-hadoop3\\spark-4.1.1-bin-hadoop3\\spark-data\\wc.txt")
val textFile: org.apache.spark.rdd.RDD[String] = D:\\Downloads\\spark-4.1.1-bin-hadoop3\\spark-4.1.1-bin-hadoop3\\spark-data\\wc.txt MapPartitionsRDD[1] at textFile at <console>:1

scala> val counts = textFile.flatMap(line => line.split(" ")).map(word => (word, 1)).reduceByKey(_ + _)
val counts: org.apache.spark.rdd.RDD[(String, Int)] = ShuffledRDD[4] at reduceByKey at <console>:1

scala> val sorted = counts.collect().sortBy(_._2)
val sorted: Array[(String, Int)] = Array((spark,6), (hello,4), (java,3), (python,2), (world,1))

scala> sorted.foreach(println)
(spark,6)
(hello,4)
(java,3)
(python,2)
(world,1)

scala> for((k,v) <- sorted) {
  |   if(v > 4) {
  |     print(k + " -> ")
  |     print(v)
  |     println()
  |   }
  | }
spark -> 6

scala> sorted.filter(_._2 > 4).foreach(println)
(spark,6)
```

PROGRAM – 10

Write a simple streaming program in Spark to receive text data streams on a particular port, perform basic text cleaning (like white space removal, stop words removal, lemmatization, etc.), and print the cleaned text on the screen. (Open Ended Question).

```
c:\Users\91988>spark-shell
WARNING: Using incubator modules: jdk.incubator.vector
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
Welcome to

  /\_/\
 /\_/\  version 4.1.1
 /\_/\

Using Scala version 2.13.17 (OpenJDK 64-Bit Server VM, Java 17.0.18)
Type in expressions to have them evaluated.
Type :help for more information.
Spark context Web UI available at http://LAPTOP-6DEICVVQ:4040
Spark context available as 'sc' (master = local[*], app id = local-1777009385296).
Spark session available as 'spark'.

scala> import spark.implicits._
import spark.implicits._

scala> spark.sparkContext.setLogLevel("ERROR")

scala> val lines = spark.readStream.format("socket").option("host", "localhost").option("port", 9999).load()
val lines: org.apache.spark.sql.DataFrame = [value: string]

scala> val words = lines.as[String]
val words: org.apache.spark.sql.Dataset[String] = [value: string]

scala> val stopWords = Seq("is", "the", "a", "an", "and", "to")
val stopWords: Seq[String] = List(is, the, a, an, and, to)
```

```
scala> val cleaned = words.map { line =>
  |   val lower = line.toLowerCase
  |   val noSpace = lower.trim.replaceAll("\\s+", " ")
  |   val noPunct = noSpace.replaceAll("[^a-zA-Z ]", "")
  |   val filtered = noPunct.split(" ").filter(word => !stopWords.contains(word))
  |   val lemmatized = filtered.map {
  |     case w if w.endsWith("ing") => w.dropRight(3)
  |     case w if w.endsWith("ed")  => w.dropRight(2)
  |     case w => w
  |   }
  |   lemmatized.mkString(" ")
  | }
val cleaned: org.apache.spark.sql.Dataset[String] = [value: string]

scala> val query = cleaned.writeStream.outputMode("append").format("console").option("checkpointLocation", "C:/tmp/check
point").start()
val query: org.apache.spark.sql.streaming.StreamingQuery = org.apache.spark.sql.execution.streaming.runtime.StreamingQueryWrapper@53f6a24

-----
Batch: 0
-----
+-----+
|value|
+-----+
-----

Batch: 1
-----
+-----+
|      value      |
+-----+
|she danc sing song|
+-----+
```

```
Batch: 2
-----
+-----+
|           value|
+-----+
|cats are play in ...|
+-----+

Batch: 3
-----
+-----+
|           value|
+-----+
|he walk talk teacher|
+-----+
```